

CASE STUDY

BANANA PEEL CEMENT ADDITIVE

Dried, milled banana peel added at just **0.2% of the binder** raised 28-day mortar strength to **61.8 MPa**, about **18% higher** than the control, while improving workability. An everyday agricultural waste became a working cementitious ingredient.

Turning Banana Peel Waste Into a Cement Additive

Cement production accounts for roughly 7% to 8% of global CO₂ emissions, and the usual supplementary materials such as fly ash and silica fume are scarce in many regions. Meanwhile the banana industry discards more than 40 million tonnes of peel every year, which rots into methane. This work, carried out at the University of Arizona, tested whether that peel could be processed into a powder that strengthens cement-based materials.

OVERVIEW

Banana peel powder (PBP) was blended into mortar at 0%, 0.2%, 0.6% and 1.25% of the binder weight, then tested for flow, setting time, compressive strength and microstructure.

The result was a clear sweet spot: a very small dose made mortar stronger and easier to place, while larger doses slowed setting and cut strength. Finding that window quickly is exactly the kind of screening Better Materials automates.

THE PROJECT

Peels were sun-dried, oven-dried, disc-milled and sieved to cement fineness, then mixed with Type II/V Portland cement, natural sand and water and cast into 50 mm cubes.

The study was built into engineering coursework: students prepared the powder, mixed and cast mortar, and ran flow, Vicat, strength, XRD and SEM tests, connecting their data to sustainable-development goals.

At a Glance

- **Feedstock:** banana peel waste, dried and milled below 75 µm.
- **Dosages:** 0%, 0.2%, 0.6% and 1.25% of binder weight.
- **Binder:** Type II/V Portland cement with natural sand.
- **Tests:** flow table, Vicat setting time and strength at 7, 14 and 28 days.
- **Characterisation:** XRD and electron microscopy at 28 days.



Valorise Agro-Waste

Turn a methane-producing waste into a useful construction ingredient.



Find the Right Dose

Pinpoint the dosage that helps performance before it starts to hurt.



Prove the Mechanism

Confirm with XRD and SEM why the best mix performs better.

The Challenge

Banana peel is mostly organic: cellulose, hemicellulose and lignin make up 40% to 50% of its dry weight, alongside 10% to 18% silica and a lot of potassium. Some of that chemistry can help cement hydrate; too much organic matter and sugars can coat cement grains and stall the reaction.

The task was to process the peel consistently and find the dosage where the benefits win.

01

Sun-dry

Peels dried in natural sunlight (about 30 °C) for 72 hours in a dust-free area.

02

Oven-dry

A further 24 hours at 65 °C to remove remaining moisture and reach constant mass.

03

Mill

Ground in a disc mill at 500 rpm for 10 minutes for a uniform powder.

04

Sieve and test

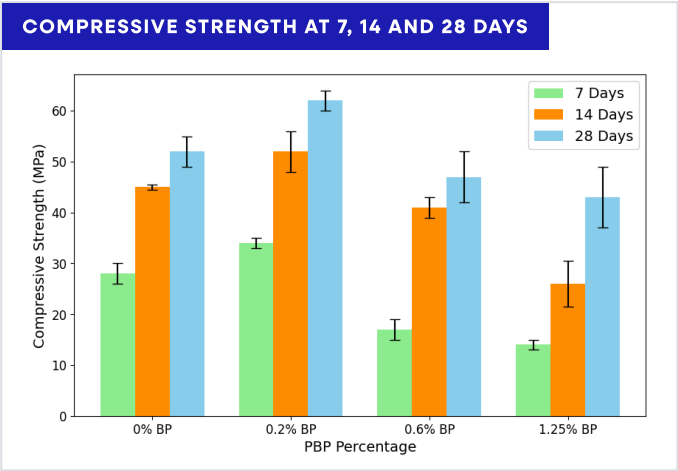
Passed through a 75 µm sieve to match cement fineness, then blended into mortar and tested.

What It Means for Manufacturers

- Local agricultural waste can become a performance additive, not just a filler.
- Dosage matters: the right window is narrow, so screen it systematically.
- Extra strength gives headroom to rebalance mixes for cost and carbon.

Results

At 0.2%, the banana peel mix was the strongest at every age: 35.9 MPa at 7 days, 50.7 MPa at 14 days and 61.8 MPa at 28 days, against 52.4 MPa for the control. Higher doses fell behind, reaching 49.5 MPa (0.6%) and 44.1 MPa (1.25%) at 28 days.



Workability

Small additions made the mortar flow more easily, with the peak at 0.6%. At 1.25% the fibrous powder began absorbing free water, but flow stayed above the control.

165

mm · 0%

172

mm · 0.2%

185

mm · 0.6%

175

mm · 1.25%

28-day strength at 0.2%

61.8

MPa, vs. 52.4 MPa control

Strength gain

+18%

from a 0.2% dose

Built Into the Classroom

The research doubled as experiential learning for engineering students, turning a lab study into hands-on sustainability training.

Hands-On Lab Work

Students processed the peel, mixed and cast mortar, and ran flow, Vicat, strength, XRD and SEM tests themselves.

Real Data, Real Analysis

Lab reports and reflections linked their own results to the environmental cost of cement.

Circular Thinking

Work tied to UN Sustainable Development Goals 9 and 12: industry innovation and responsible production.

Results in Detail

Setting behaviour, mineralogy and microstructure all point the same way: a small dose of banana peel powder works with cement hydration, while larger doses introduce enough organic material to get in the way.

